

# AI LAB 2

```
import java.util.*;

class A_star {

    static int[][] goal = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 0}
    };

    static int heuristic(int[][] state) {
        int h = 0;
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 3; j++)
                if (state[i][j] != 0 && state[i][j] != goal[i][j])
                    h++;
        return h;
    }

    static void print(int[][] s) {
        for (int[] row : s)
            System.out.println(Arrays.toString(row));
        System.out.println();
    }

    public static void main(String[] args) {

        int[][] start = {
            {1, 2, 3},
            {4, 0, 6},
            {7, 5, 8}
        };

        Queue<int[][]> q = new LinkedList<>();
        Set<String> visited = new HashSet<>();
    }
}
```

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q.add(start);

while (!q.isEmpty()) {

    int[][] curr = q.poll();
    print(curr);

    if (Arrays.deepEquals(curr, goal)) {
        System.out.println("Goal Reached");
        return;
    }

    visited.add(Arrays.deepToString(curr));

    int x = 0, y = 0;

    for (int i = 0; i < 3; i++)
        for (int j = 0; j < 3; j++)
            if (curr[i][j] == 0) {
                x = i;
                y = j;
            }

    int[][] dir = {
        {1, 0},
        {-1, 0},
        {0, 1},
        {0, -1}
    };

    for (int[] d : dir) {

        int nx = x + d[0];
        int ny = y + d[1];

        if (nx >= 0 && ny >= 0 && nx < 3 && ny < 3) {

            int[][] newState = new int[3][3];

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        for (int i = 0; i < 3; i++)
            newState[i] = curr[i].clone();

        newState[x][y] = newState[nx][ny];
        newState[nx][ny] = 0;

        if (!visited.contains(Arrays.deepToString(newState))) {
            q.add(newState);
        }
    }
}

```

```
Console X
<terminated> A_star [Java Application] /Applications/Eclipse Java.app/Contents/Eclipse/plugins/org
[1, 2, 3]
[4, 0, 6]
[7, 5, 8]

[1, 2, 3]
[4, 5, 6]
[7, 0, 8]

[1, 0, 3]
[4, 2, 6]
[7, 5, 8]

[1, 2, 3]
[4, 6, 0]
[7, 5, 8]

[1, 2, 3]
[0, 4, 6]
[7, 5, 8]

[1, 2, 3]
[4, 5, 6]
[7, 8, 0]

Goal Reached
```